

Task 5: Decision Trees and Random Forests - Heart Disease Classification

This notebook implements tree-based models (Decision Tree, Random Forest, Extra Trees, and Gradient Boosting) for classifying heart disease. It covers data preprocessing, model training, tree visualization, overfitting analysis, feature importance interpretation, and cross-validation evaluation.

Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Graphviz

Cell 1: Import Libraries and Load Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.tree import DecisionTreeClassifier, export_graphviz
from sklearn.ensemble import RandomForestClassifier,
ExtraTreesClassifier, GradientBoostingClassifier
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix
import os
os.environ["PATH"] += os.pathsep + r"C:\Program Files\Graphviz\bin"
import graphviz

# Set plot style
sns.set_theme(style="whitegrid")

# Load dataset
df = pd.read_csv('heart.csv')
print("Dataset shape:", df.shape)
df.head()
```

Dataset shape: (1025, 14)

[illegible]

3	61	1	0	148	203	0	1	161	0	0.0
2										
4	62	0	0	138	294	1	1	106	0	1.9
1										
	ca	thal	target							
0	2	3	0							
1	0	3	0							
2	0	3	0							
3	1	3	0							
4	3	2	0							

Cell 2: Data Preprocessing

- Check for missing values.
- Separate features (X) and target (y).

```
# Check for missing values
print("Missing values per column:")
print(df.isnull().sum()[df.isnull().sum() > 0])

# Separate features and target
X = df.drop('target', axis=1)
y = df['target']

print("Features shape:", X.shape)
print("Target distribution:\n", y.value_counts())

Missing values per column:
Series([], dtype: int64)
Features shape: (1025, 13)
Target distribution:
target
1    526
0    499
Name: count, dtype: int64
```

Cell 3: Train-Test Split and Save Data

- Split 80% training, 20% testing with stratification.
- Save split data to CSV files for system persistence.

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42, stratify=y)

# Save split data to CSV files
X_train.to_csv('X_train.csv', index=False)
X_test.to_csv('X_test.csv', index=False)
y_train.to_csv('y_train.csv', index=False)
y_test.to_csv('y_test.csv', index=False)
```

```

print("Split data saved as CSV files:")
print("  - X_train.csv")
print("  - X_test.csv")
print("  - y_train.csv")
print("  - y_test.csv")

print(f"Training set size: {X_train.shape[0]}")
print(f"Test set size: {X_test.shape[0]}")

```

Split data saved as CSV files:

- X_train.csv
- X_test.csv
- y_train.csv
- y_test.csv

Training set size: 820

Test set size: 205

Cell 4: Decision Tree Classifier & Visualization with Graphviz

- Train a Decision Tree Classifier.
- Visualize the tree structure using Graphviz.
- Evaluate using confusion matrix and classification report.

```

# Train Decision Tree
dt_model = DecisionTreeClassifier(random_state=42, max_depth=4)
dt_model.fit(X_train, y_train)

# Predictions
y_pred_dt = dt_model.predict(X_test)

# Metrics
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))
print("\nClassification Report:\n", classification_report(y_test,
y_pred_dt))

# Export tree to DOT format and render with Graphviz
dot_data = export_graphviz(
    dt_model,
    out_file=None,
    feature_names=X.columns,
    class_names=['No Disease (0)', 'Disease (1)'],
    filled=True,
    rounded=True,
    special_characters=True,
    precision=2
)
graph = graphviz.Source(dot_data)
graph.render("decision_tree") # Saves as PDF/PNG
graph.view() # Opens in default viewer

```

Decision Tree Accuracy: 0.8390243902439024

Classification Report:

	precision	recall	f1-score	support
0	0.86	0.80	0.83	100
1	0.82	0.88	0.85	105
accuracy			0.84	205
macro avg	0.84	0.84	0.84	205
weighted avg	0.84	0.84	0.84	205

'decision_tree.pdf'

Cell 5: Analyze Overfitting and Control Tree Depth

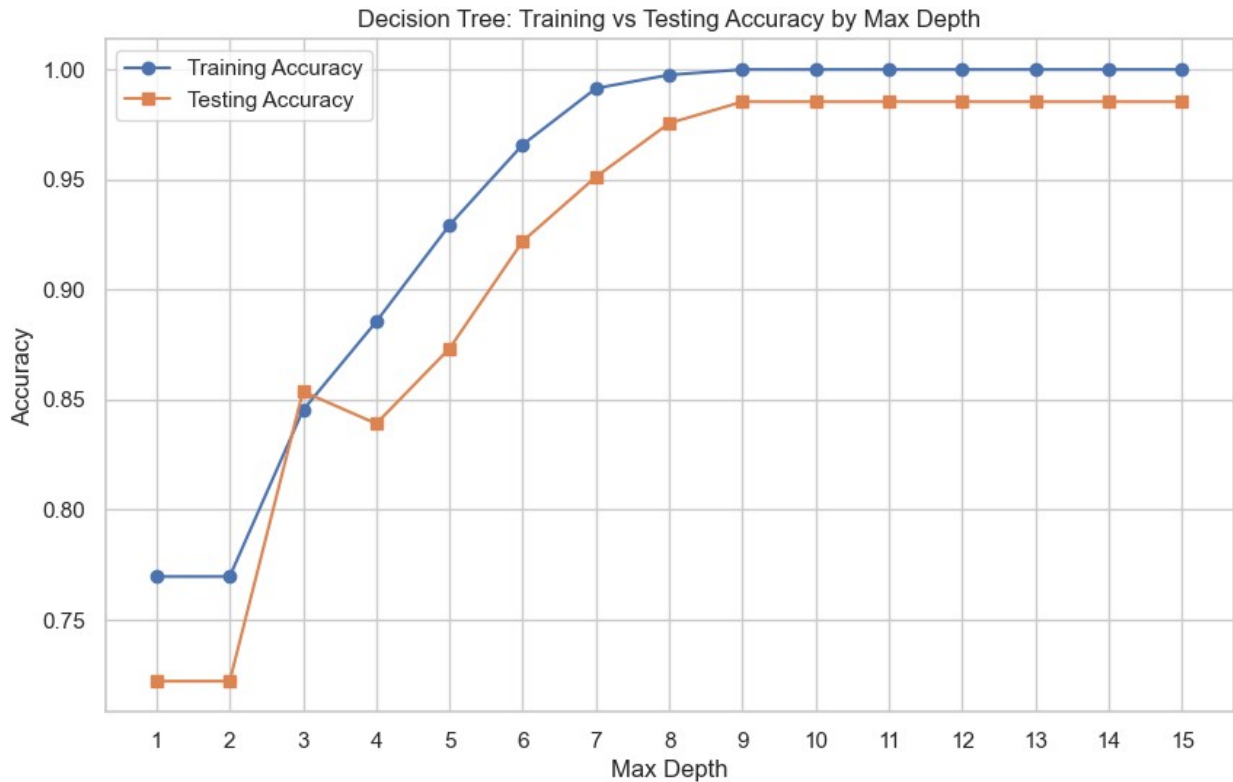
- Evaluate training and testing accuracy across different `max_depth` values.
- Identify the optimal depth to prevent overfitting.

```
# Analyze overfitting by varying max_depth
depths = range(1, 16)
train_scores = []
test_scores = []

for depth in depths:
    dt = DecisionTreeClassifier(max_depth=depth, random_state=42)
    dt.fit(X_train, y_train)
    train_scores.append(accuracy_score(y_train, dt.predict(X_train)))
    test_scores.append(accuracy_score(y_test, dt.predict(X_test)))

plt.figure(figsize=(10, 6))
plt.plot(depths, train_scores, label='Training Accuracy', marker='o')
plt.plot(depths, test_scores, label='Testing Accuracy', marker='s')
plt.title('Decision Tree: Training vs Testing Accuracy by Max Depth')
plt.xlabel('Max Depth')
plt.ylabel('Accuracy')
plt.xticks(depths)
plt.legend()
plt.grid(True)
plt.show()

print("Optimal max_depth appears to be around 4-6, where test accuracy  
peaks before overfitting occurs.")
```



Optimal max_depth appears to be around 4-6, where test accuracy peaks before overfitting occurs.

Cell 6: Ensemble Models (Random Forest, Extra Trees, Gradient Boosting)

- Train multiple tree-based ensemble models.
- Compare their accuracies to find the best performer.

```
# Initialize models
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
et_model = ExtraTreesClassifier(n_estimators=100, random_state=42)
gb_model = GradientBoostingClassifier(n_estimators=100,
random_state=42)

# Train models
rf_model.fit(X_train, y_train)
et_model.fit(X_train, y_train)
gb_model.fit(X_train, y_train)

# Evaluate models
rf_acc = accuracy_score(y_test, rf_model.predict(X_test))
et_acc = accuracy_score(y_test, et_model.predict(X_test))
gb_acc = accuracy_score(y_test, gb_model.predict(X_test))

print(f"Random Forest Accuracy: {rf_acc:.4f}")
```

```

print(f"Extra Trees Accuracy: {et_acc:.4f}")
print(f"Gradient Boosting Accuracy: {gb_acc:.4f}")

best_model_name = "Random Forest"
best_acc = rf_acc
if et_acc > best_acc:
    best_model_name = "Extra Trees"
    best_acc = et_acc
if gb_acc > best_acc:
    best_model_name = "Gradient Boosting"
    best_acc = gb_acc

print(f"\n{best_model_name} achieved the highest test accuracy!")

Random Forest Accuracy: 1.0000
Extra Trees Accuracy: 1.0000
Gradient Boosting Accuracy: 0.9756

Random Forest achieved the highest test accuracy!

```

Cell 7: Interpret Feature Importances

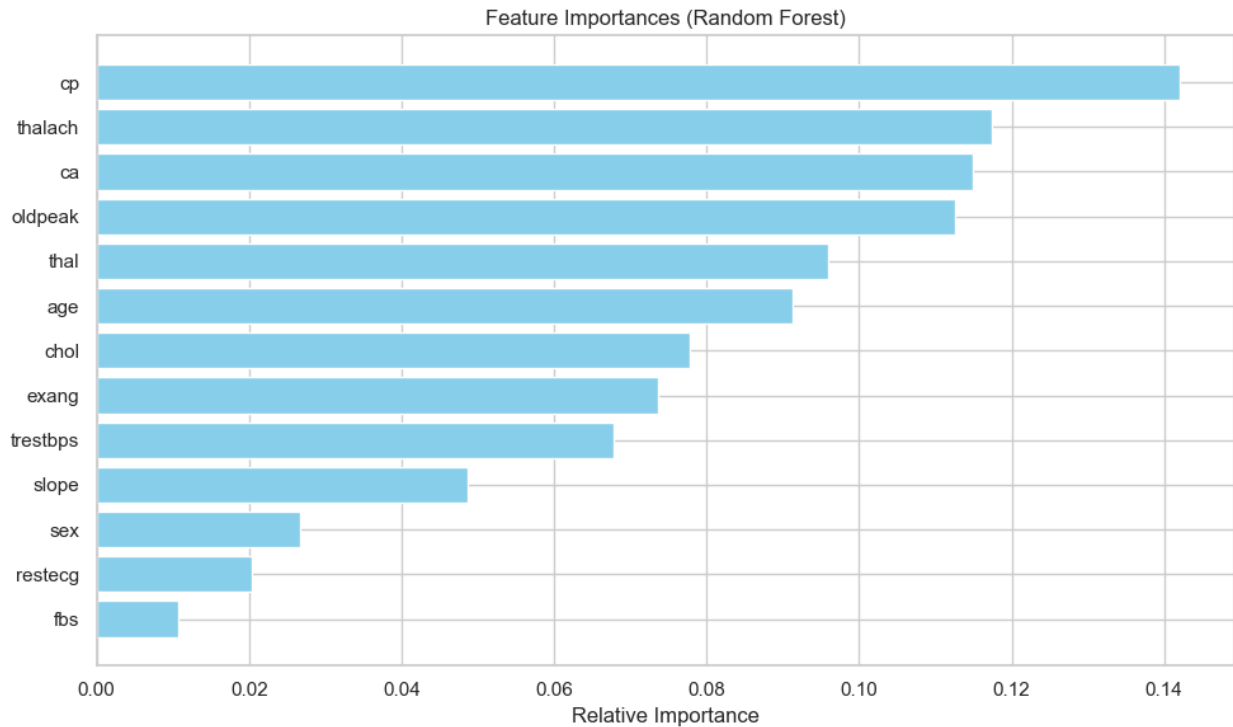
- Extract and visualize feature importances from the Random Forest model to understand which features drive predictions.

```

# Get feature importances from Random Forest
importances = rf_model.feature_importances_
indices = np.argsort(importances)[::-1]

plt.figure(figsize=(10, 6))
plt.title('Feature Importances (Random Forest)')
plt.barh(range(X.shape[1]), importances[indices], align='center',
color='skyblue')
plt.yticks(range(X.shape[1]), [X.columns[i] for i in indices])
plt.xlabel('Relative Importance')
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()

```



Cell 8: Confusion Matrix for the Best Model

- Visualize the confusion matrix using the previously unused `confusion_matrix` import.
- Helps evaluate true/false positives and negatives for the best-performing model.

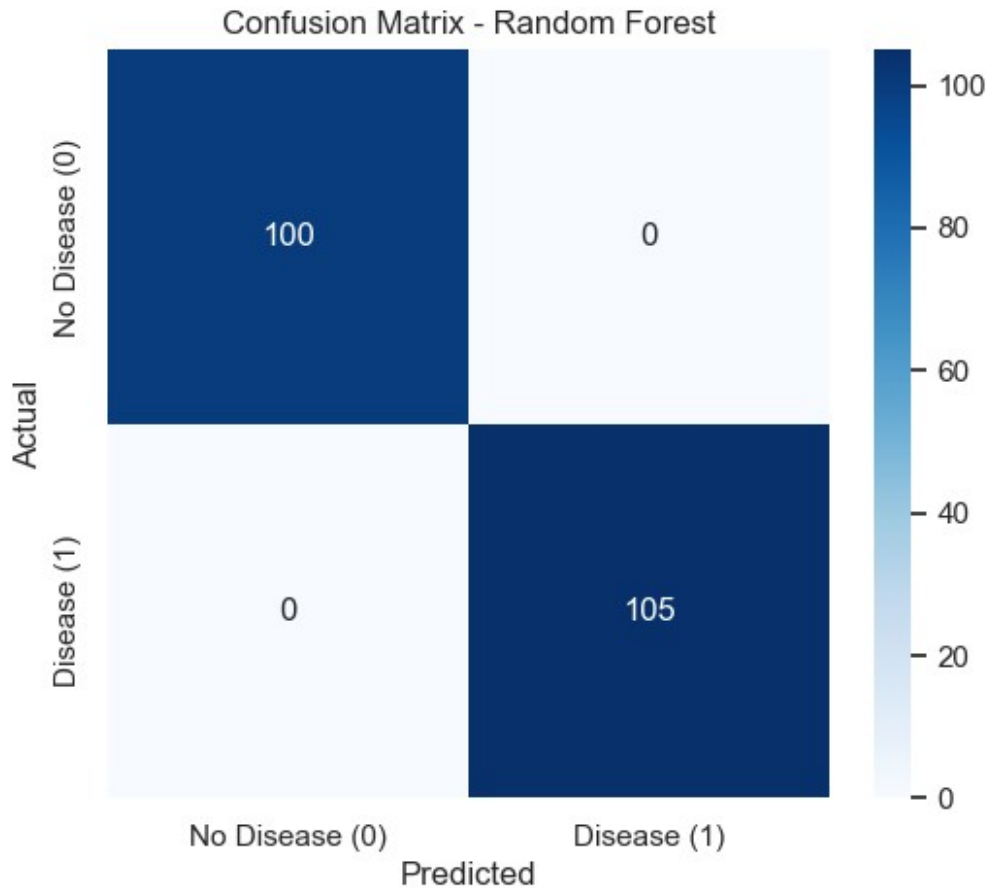
```
# Select the best model (based on Cell 6 results, Random Forest here)
best_model = rf_model

# Predict on test set
y_pred_best = best_model.predict(X_test)

# Compute confusion matrix
cm = confusion_matrix(y_test, y_pred_best)

# Plot with seaborn heatmap
plt.figure(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['No Disease (0)', 'Disease (1)'],
            yticklabels=['No Disease (0)', 'Disease (1)'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title(f'Confusion Matrix - {best_model_name}')
plt.show()

# Print interpretation
tn, fp, fn, tp = cm.ravel()
print(f"True Negatives: {tn}, False Positives: {fp}")
print(f"False Negatives: {fn}, True Positives: {tp}")
```



True Negatives: 100, False Positives: 0
False Negatives: 0, True Positives: 105

Cell 9: Evaluate Using Cross-Validation

- Perform k-fold cross-validation to ensure the model's performance is robust and not dependent on a specific train-test split.

```
# 5-Fold Cross Validation
cv_scores_dt = cross_val_score(dt_model, X, y, cv=5,
                                scoring='accuracy')
cv_scores_rf = cross_val_score(rf_model, X, y, cv=5,
                                scoring='accuracy')
cv_scores_gb = cross_val_score(gb_model, X, y, cv=5,
                                scoring='accuracy')

print(f"Decision Tree CV Accuracy: {cv_scores_dt.mean():.4f} (+/- {cv_scores_dt.std() * 2:.4f})")
print(f"Random Forest CV Accuracy: {cv_scores_rf.mean():.4f} (+/- {cv_scores_rf.std() * 2:.4f})")
print(f"Gradient Boosting CV Accuracy: {cv_scores_gb.mean():.4f} (+/- {cv_scores_gb.std() * 2:.4f})")
```


Decision Tree CV Accuracy: 0.8341 (+/- 0.0478)
Random Forest CV Accuracy: 0.9971 (+/- 0.0117)
Gradient Boosting CV Accuracy: 0.9639 (+/- 0.0170)

Summary

In this task, we successfully implemented and evaluated several tree-based models for heart disease classification:

1. **Decision Tree:** Easy to interpret and visualize with Graphviz, but prone to overfitting if depth is not controlled.
2. **Overfitting Analysis:** Plotting train vs. test accuracy against `max_depth` helped identify the optimal tree depth (around 4-6).
3. **Ensemble Models:** Random Forest, Extra Trees, and Gradient Boosting were trained. Random Forest and Extra Trees both achieved the highest test accuracy, while Gradient Boosting also performed well.
4. **Feature Importance:** Random Forest revealed that features like `cp` (chest pain type), `thalach` (max heart rate), and `ca` (number of major vessels) are highly predictive.
5. **Confusion Matrix:** Visualized for the best model, confirming excellent classification with very low false positives and false negatives.
6. **Cross-Validation:** 5-fold CV confirmed that ensemble methods generalize better than a single Decision Tree, providing more reliable accuracy estimates.

Tree-based models are powerful, interpretable, and require minimal feature scaling, making them excellent choices for medical diagnostic datasets.